DeLP3E Approximation Experiment

New Features Reference

Definitions for the network-, centrality- and ratio-based features added on top of the original dataset.
These are the predictors used by the abstract decision tree.

5 sections · 38 feature entries

Generated by sampling/generate_new_feature_docs.py

EM — Bayesian Network graph metrics

em_graph_density
float in [0, 1]

Density of the directed BN graph: ratio of present arcs to all possible arcs. A measure of how interconnected the variables are.
How computed: networkx.density(G) where G is the directed BN graph.

em_clustering_coef
float in [0, 1]

Average local clustering coefficient on the undirected version of the BN. Captures the tendency of variables to form triangles (common-cause / common-effect structures).
How computed: networkx.average_clustering(G.to_undirected()).

em_avg_shortest_path
float >= 0

Average shortest path length over the largest connected component of the BN. Indicates how 'reachable' BN variables are from each other on average.
How computed: networkx.average_shortest_path_length on the largest CC.

em_diameter
int >= 0

Diameter (longest shortest path) of the largest connected component of the BN.
How computed: networkx.diameter on the largest CC.

em_deg_centrality_avg / em_deg_centrality_max / em_deg_centrality_std
three floats in [0, 1]

Aggregate of normalized degree centrality (in+out) over all BN nodes. The std measures how heterogeneous the connectivity is.
How computed: networkx.degree_centrality(G); then mean / max / standard deviation.

em_closeness_avg / em_closeness_max
floats in [0, 1]

Mean and maximum closeness centrality: how close (inverse path length) a node is to all others. Nodes with high closeness propagate evidence quickly through the network.
How computed: networkx.closeness_centrality on the undirected version.

em_betweenness_avg / em_betweenness_max
floats in [0, 1]

Mean and maximum betweenness centrality: the fraction of shortest paths that pass through a node. Captures BN 'bottlenecks'.
How computed: networkx.betweenness_centrality(G).

em_eigenvector_avg / em_eigenvector_max
floats in [0, 1]

Mean and maximum eigenvector centrality: a node is important if its neighbours are important. Highlights influential BN variables.
How computed: networkx.eigenvector_centrality_numpy(G); falls back to power iteration on the undirected graph if not convergent.

em_in_degree_gini
float in [0, 1]

Gini coefficient of the in-degree distribution. 0 means every node has the same number of parents; 1 means one node concentrates all the in-degree.
How computed: Gini coefficient applied to the list of in-degrees.

AM — Attack-graph metrics (dialectical)

am_attack_density
float in [0, 1]

Density of the AM attack graph (arguments as nodes, defeats as directed edges).
How computed: networkx.density on the attack graph parsed from the solver's dGraph output.

am_attack_clustering
float in [0, 1]

Average clustering coefficient of the (undirected) attack graph. Higher values indicate clusters of mutually attacking arguments.
How computed: networkx.average_clustering on the undirected attack graph.

am_attack_avg_path
float >= 0

Average shortest path length in the largest connected component of the attack graph.
How computed: networkx.average_shortest_path_length on the largest CC.

am_attack_diameter
int >= 0

Diameter of the largest connected component of the attack graph.
How computed: networkx.diameter on the largest CC.

am_attack_reciprocity
float in [0, 1]

Fraction of attack edges (A->B) that are reciprocated (B->A). Measures how often arguments mutually attack each other.
How computed: networkx.reciprocity on the directed attack graph.

am_attack_deg_avg / max / std
three floats in [0, 1]

Aggregates of degree centrality (in+out) on the attack graph. High values flag arguments that participate in many defeats.
How computed: networkx.degree_centrality; mean / max / std.

am_attack_closeness_avg / max
floats in [0, 1]

Average and max closeness centrality on the attack graph.
How computed: networkx.closeness_centrality on the undirected attack graph.

am_attack_betweenness_avg / max
floats in [0, 1]

Average and max betweenness centrality on the attack graph: identifies 'bridging' arguments through which many defeat chains pass.
How computed: networkx.betweenness_centrality on the directed attack graph.

am_attack_eigenvector_avg / max
floats in [0, 1]

Average and max eigenvector centrality on the attack graph: an argument is influential if it attacks/is attacked by influential arguments.
How computed: networkx.eigenvector_centrality_numpy on the directed attack graph.

am_attack_in_degree_gini
float in [0, 1]

Gini coefficient of the in-degree (number of attackers) distribution in the attack graph. High values indicate a few heavily attacked 'target' arguments.
How computed: Gini coefficient applied to the in-degree list.

AF — Bipartite annotation graph

af_bipartite_density
float in [0, 1]

Density of the bipartite graph whose two parts are annotated rules and BN variables, with an edge whenever a rule's annotation references a variable. Captures how much of the rule x variable matrix is actually populated.
How computed: total_edges / (n_annotated_rules * em_var).

af_var_usage_gini
float in [0, 1]

Gini coefficient of how often each BN variable appears across annotations. 0 = all BN variables used equally; 1 = annotations rely on a single variable.
How computed: Gini coefficient over var_usage counts.

af_var_coverage
float in [0, 1]

Fraction of BN variables that appear in at least one annotation formula.
How computed: n_distinct_vars_in_annotations / em_var.

af_avg_var_usage / af_max_var_usage
float, int

Average and maximum number of annotations that reference a given BN variable. Indicates whether annotations spread the BN load or concentrate it.
How computed: mean / max over the variable-usage histogram.

af_rule_to_var_ratio
float >= 0

Number of annotated rules per BN variable used. A high value means many annotations share a small set of BN variables.
How computed: n_annotated_rules / n_distinct_vars_in_annotations.

Derived scale-invariant ratios (used by the abstract tree)

em_density
float >= 0

Arcs per BN node. Approximates the average degree of the network.
How computed: em_n_arcs / em_var.

em_entropy_per_var
float in [0, 1] (typically)

Joint entropy of the BN divided by the number of variables. Independent of network size: low values mean concentrated mass in a few worlds; values close to 1 mean an approximately uniform distribution.
How computed: em_entropy / em_var.

em_treewidth_ratio
float in [0, 1]

Treewidth relative to network size. Captures how 'inferentially hard' the BN is regardless of absolute size.
How computed: em_treewidth / em_var.

em_max_indeg_ratio
float in [0, 1]

Maximum number of parents observed for any node, normalized by the total number of nodes.
How computed: em_max_in_degree / em_var.

am_arg_density
float >= 0

Average number of arguments per rule. Captures how 'productive' the rule base is in generating arguments.
How computed: am_n_arguments / n_rules.

am_defeat_ratio
float >= 0

Average number of defeats per argument. Captures the conflict intensity inside the AM.
How computed: am_n_defeaters / am_n_arguments.

am_tree_density
float in [0, 1]

Average number of dialectical trees per argument.
How computed: am_n_trees / am_n_arguments.

af_em_coverage
float in [0, 1]

Fraction of BN variables that are referenced by at least one annotation formula. Identical in interpretation to af_var_coverage; kept as a redundant alias for legacy code paths.
How computed: af_n_em_vars_used / em_var.

lit_head_def_rate
float in [0, 1]

Number of defeasible rules where the queried literal is the head, divided by the total number of rules in the AF.
How computed: lit_head_def / n_rules.

lit_head_strict_rate
float in [0, 1]

Same as above but for strict rules.
How computed: lit_head_strict / n_rules.

Derived scale-invariant ratios (used by the abstract tree) (continued)

lit_body_rate
float in [0, 1]

Number of rules where the queried literal appears in the body, normalized by n_rules.
How computed: lit_body_count / n_rules.

lit_complement_body_rate
float in [0, 1]

Same as above but for the literal's complement.
How computed: lit_complement_body / n_rules.

lit_complement_head_rate
float in [0, 1]

Same as above but for the complement appearing as head.
How computed: lit_complement_head / n_rules.

Sign / direction features

space_orientation

categorical in {worlds-favored, balanced, programs-favored}

Discretization of annots_minus_em. Captures the direction of the search-space asymmetry while throwing away its absolute magnitude — the magnitude is dataset-specific, the sign is not.

How computed: +1 if annots_minus_em > 1; -1 if < -1; 0 otherwise.